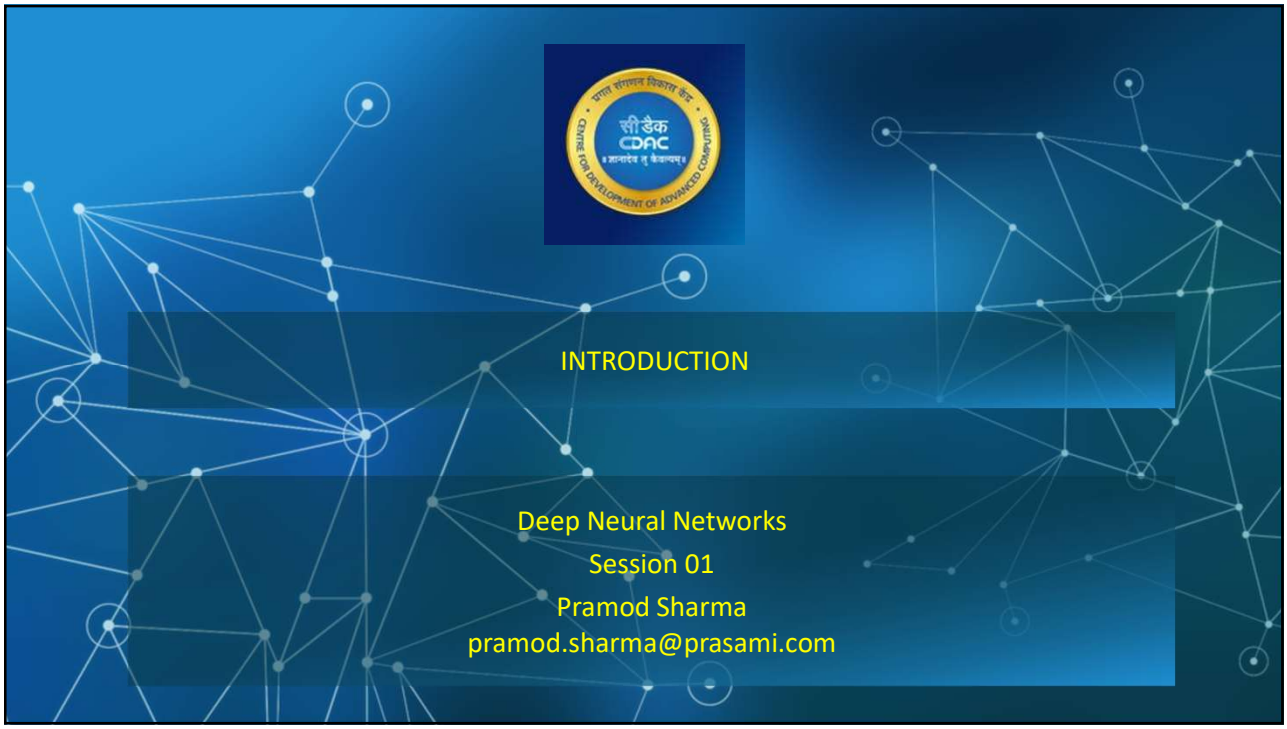

INTRODUCTION
Deep Neural Networks
Session 01
Pramod Sharma
pramod.sharma@prasami.com


2
Agenda
Introduction
AI vs ML vs Deep Learning
MP Neuron
5/2/2026
pra-sami

3

References

- Deep Learning using Python, S Lovelyn Rose,L Ashok Kumar, D Karthika R/ Wiley India, 1st Edition, 9788126579914
- Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Neural Networks and Learning Machines, Simon Haykin
- Pattern Recognition and Machine Learning, Christopher M. Bishop
- Deep Learning with Python - François Chollet
- Hands-On Machine Learning with Scikit-Learn and TensorFlow
- TensorFlow Deep Learning Cookbook

pra-sâmi

4

"*Theory exam– 40% weightage*

Lab exam – 40% weightage

Internal exam – 20% weightage"

Evaluation method

pra-sâmi

5

Logistics



"The man who asks question is a fool for a minute, a man who doesn't ask is a fool for life."
~ Confucius

pra-sâmi

6

Logistics

- We encourage 'discussion groups':
  - Study groups
  - Whatsapp groups
- Expect you to complete your assignment individually!
- No group assignments unless stated otherwise
- Code is small part of it
- Pay special attention to inline comments
  - Comments should focus on what you were trying to implement



pra-sâmi

8 Uncertainty Everywhere



Goal:
Delivering a passenger to the airport on time



- The agent forms a plan, let's say… A90,
  - Leave home 90 minutes before the flight departs
  - Driving at a reasonable speed
- Are you certain "*Plan A90 will get us to the airport in time*."?
  - Not in absolute sense but with some riders

pra-sami

9 Uncertainty Everywhere



Goal:
Delivering a passenger to the airport on time



- How about other plans, such as A180,
  - Might increase the agent's belief that it will get to the airport on time,
  - But also increase the likelihood of a long wait
- Probability is an agent's measure of belief in some proposition — subjective probability.
- An agent's belief depends on its prior belief and what it observes.

pra-sami

10

Decision-Making Under Uncertainty

- No agent has complete information about the world
- Decision-making must account for incomplete and uncertain data
- Mere assumptions are inadequate for robust action
  - Example: Wearing a seatbelt due to accident risk
- Reasoning about uncertainty is essential for intelligent behavior
- Every action under uncertainty equates to a gamble — managed through probability theory



5/2/2026

pra-sami

11

Overview

- Nature vs. Computation
  - Real world is a continuum — smooth and seamless.
  - Mathematics, however, models reality using discr
  - Early film-based images were continuous n tal images break reality into pixels.
- Brain vs. Artificial Computat
  - The human brain is a , and massively parallel system.
  - Traditional c ntial and linear.
  - Neural the brain — but in a highly simplified and generalized form.
  - In e rk models the way the brain approaches tasks such as perception, learning, and decis

All models are wrong... some models are useful!

Stimulus → Receptors ⇄ Neural Net ⇄ Effectors → Response

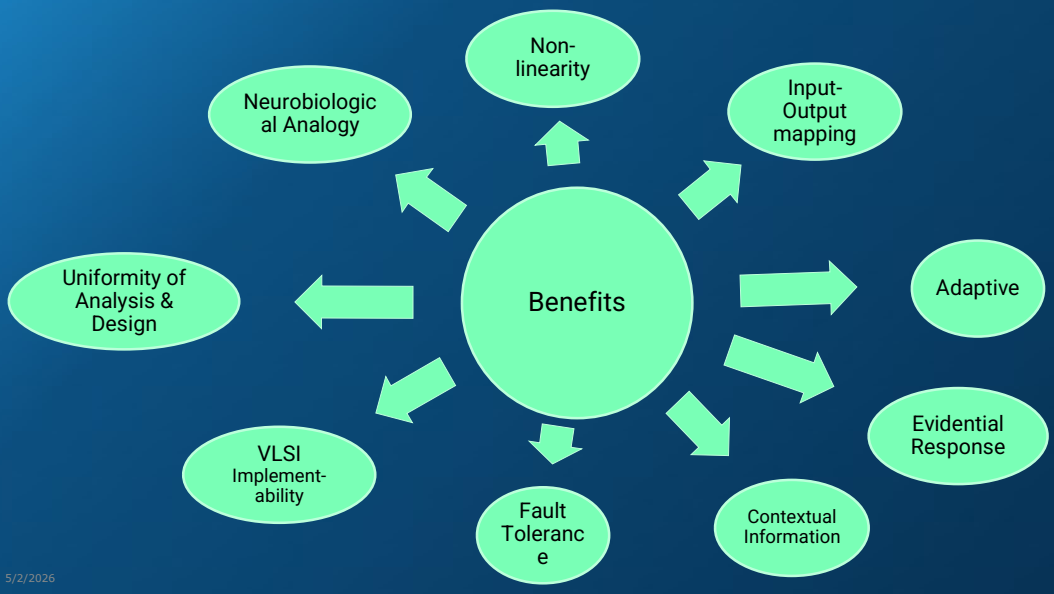
Is this how our brain works? Really!!

5/2/2026

pra-sami

12

Benefits of Neural Networks



5/2/2026

pra-sami

13

Core Capabilities of Neural Networks

- Nonlinearity
  - Can model both linear and nonlinear systems, matching real-world behavior.
- Input-Output Mapping
  - Learns complex relationships by adjusting internal weights.
- Adaptivity:
  - Quickly retrains for changing environments; handles the stability–plasticity dilemma.

5/2/2026

pra-sami

14

Advanced Functionalities of Neural Networks

Part 1

- Evidential Response
  - Provides both decision output and confidence level.
- Contextual Information
  - Knowledge embedded in structure and activation.
- Fault Tolerance
  - Continues functioning even when some neurons fail.

pra-sami

15

Advanced Functionalities of Neural Networks

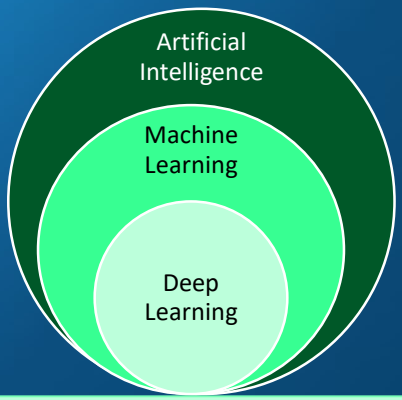
Part 2

- VLSI Implementability
  - Captures complex behavior in hardware systems.
- Uniformity of Analysis and Design
  - Modular, universal structure across domains.
- Neurobiological Analogy
  - Inspired by the brain's massively parallel, fault-tolerant architecture.

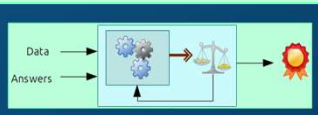
pra-sami

17

AI vs ML vs Deep Learning



AI ⊃ ML ⊃ DL (nested hierarchy)
Often used interchangeably, but technically distinct



- Artificial Intelligence (AI)
  - Broad field focused on building intelligent systems
  - Includes rule-based systems, search, planning, and learning methods
- Machine Learning (ML)
  - Systems learn patterns from historical data
  - Improves performance without explicit programming
- Deep Learning (DL)
  - Uses multi-layered neural networks (Deep Neural Networks)
  - Inspired by human brain's structure for hierarchical feature learning

5/2/2026

pra-sami

18

Artificial Intelligence vs. Machine Learning

Artificial Intelligence	Machine Learning
Broad discipline of creating intelligent systems	Subset of AI
Mimics human intelligence: reasoning, planning, problem-solving	Focuses on learning from data
Can be rule-based (no learning required)	Improves performance with experience
Can apply logic and rules to solve problems	Data-driven rather than rule-based Requires large datasets for training.
Examples: Expert systems, game-playing agents, search algorithms	Examples: Regression, classification, recommendation systems

5/2/2026

pra-sami

19

AI vs. ML - Programming and Learning

Key Differences

- Scope: AI is broader; ML is a subset
- Approach: AI may use rules; ML relies on data
- Goal: AI → intelligent behavior, ML → pattern learning
- Dependency: ML requires data; AI may not

pra-sami

20

AI vs. ML - Classification

Artificial Intelligence

- Weak AI: Task-specific (e.g., Siri, Alexa)
- General AI: Human-level intelligence (still emerging)
- Strong AI: Hypothetical, superintelligent systems

Machine Learning

- Supervised Learning: Learns from labeled data
- Unsupervised Learning: Finds patterns in unlabeled data
- Reinforcement Learning: Learns via feedback from actions

pra-sami

21 AI vs. ML - Data Dependency & Usage

Artificial Intelligence	Machine Learning
❑ Uses logic, rules, and sometimes symbolic reasoning	❑ Strongly data-driven
❑ Applied in robotics, NLP, expert systems	❑ Applied in predictive modeling, image/speech recognition, etc.
❑ Currently dominated by Weak AI applications	❑ Widely used in real-world applications
❑ General AI is under research	
❑ Strong AI is still theoretical	

5/2/2026

pra-sami

22 Deep Learning

- ❑ Large Neural Networks

"Using brain simulations, hope to:
Make learning algorithms much better and easier to use,
Make revolutionary advances in machine learning and AI,
I believe this is our best shot at progress towards real AI."

- Andrew Ng

- ❑ Learning successive layers of increasingly meaningful representations
- ❑ Modern network contain hundreds of successive layers
- ❑ Successive layers are learned via "neurons" connected via neural network

Some concepts were inspired by how our brain works
It is NOT a replica of human brain!!!

5/2/2026

pra-sami

23 Deep Learning

- Why Deep Learning is more practical today?
  - Availability of large computing power
  - Availability of large datasets
- Most flavors of the old generations of learning algorithms, performance will plateau
- Deep learning that is scalable
  - Performance just keeps getting better as more and more data is fed
- Most value today is coming from supervised learning
- Eventually, we will see benefits of unsupervised learning

pra-sami

24 Deep Learning

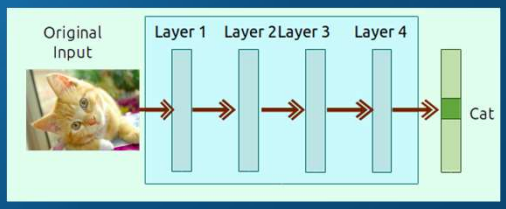
- Usually a neural network contains
  - Input Layer
  - Hidden layers [1 … n]
  - Output layer
- We may call network with 1 to 2 hidden layer as shallow
- Network with 10 or more layers as deep
  - No set demarcation!
- I guess, scientists just got excited when someone labeled them as deep network
- Intelligent software to automate routine tasks, understand speech or images, make diagnosis in medicine and support basic scientific research

pra-sami

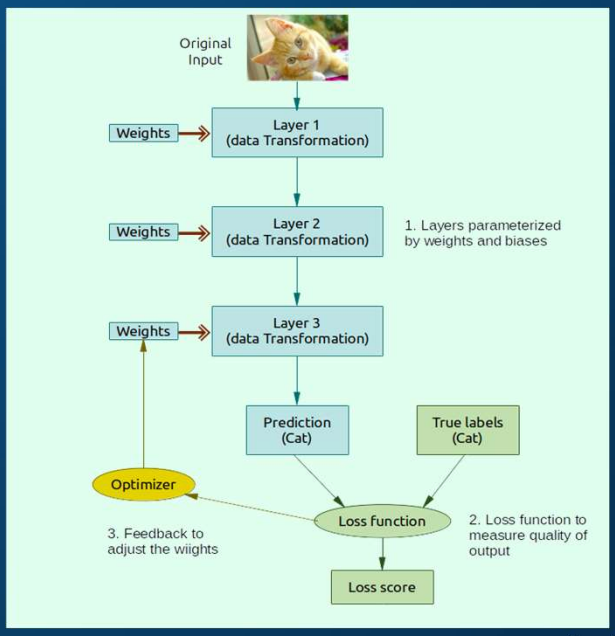
25

How deep learning works…

- As the images are processed through the layers



- The representations are increasingly filtered, purified and distilled to make them more meaningful



5/2/2026

pra-sami

26

What has been achieved so far

- Learn to see and hear… so natural to humans but elusive to machines earlier
- Image classification
- Speech recognition
- Handwriting reco
- Writing style recognition (who was the author)
- Improved machine translation
- Text-to-speech conversion
- Digital assistants such as Google Now and Amazon Alexa
- Little autonomous driving
- Improved ad targeting, as used by Google, Baidu, and Bing
- ts on the web
- Ability to answer natural-language questions
- Superhuman games playing: chess, go…

Still long way to go…
Human-level general intelligence too far away…

5/2/2026

pra-sami

27

Neurons

5/2/2026

pra-sâmi

28

To Play or Not to Play!

id	Dry Weather	Low Temp	Homework Done	Team Members	Equipment	Ground	Played
1	1	1	1	1	0	1	1
2	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1
4	0	1	0	1	1	1	0
5	0	0	1	1	1	0	0
6	0	0	0	0	0	1	0

5/2/2026

pra-sâmi

29 Neurons

- Features:
  - Is it raining?
  - Is it too hot?
  - Have I completed my homework?
  - Are sufficient players ready?
  - Is cricket equipment ready?
  - Is ground available?
- Depending on the feature values, you may get to play or not
- Features like homework and availability of ground can be considered as 'inhibitory'.

id	Dry Weather	Low Temp	Homework Done	Team Members	Equipment	Ground	Played
1	1	1	1	1	0	1	1
2	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1
4	0	1	0	1	1	1	0
5	0	0	1	1	1	0	0
6	0	0	0	0	0	1	0

- Notes :
  - Aggregator function is sum and threshold can be 3.
  - Assign 0 or 1 if a parameter is in favor or not

Given sufficient data point, we can train an algorithm to make such simple decisions for us.

5/2/2026

pra-sami

30 MP Neuron

- In 1943 Warren S. McCulloch, a neuroscientist, and Walter Pitts, a logician, published "A logical calculus of the ideas immanent in nervous activity" in the Bulletin of Mathematical Biophysics
- In this paper McCulloch and Pitts tried to understand how the brain could produce highly complex patterns by using many basic cells that are connected together
- These basic brain cells are called neurons, and McCulloch and Pitts gave a highly simplified model of a neuron in their paper
- The McCulloch and Pitts model of a neuron, which we will call an MCP neuron for short, has made an important contribution to the development of artificial neural networks -- which model key features of biological neurons

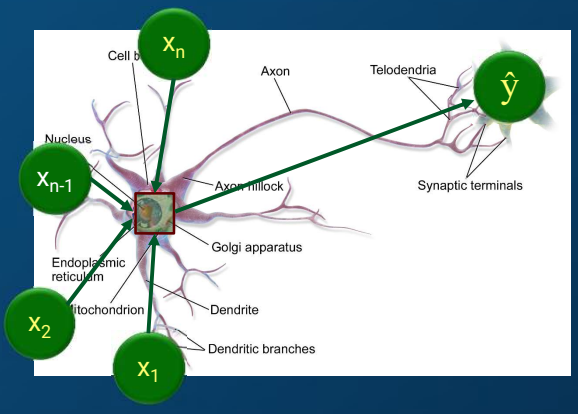
5/2/2026

pra-sami

31

McCulloch–Pitts (MP) Neuron Model

- Neurons receive inputs and produce a binary output
- Inputs are binary: $x_i \in \{0,1\}$
- Inputs can be:
  - Excitatory (contribute positively)
  - Inhibitory (dominant effect)
- If any inhibitory input is active → output = 0



pra-sami

32

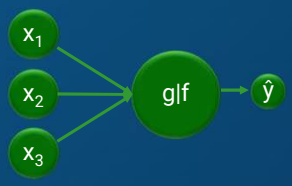
MP Neuron Computation

- Aggregation function g: sum of excitatory inputs
- Compare with threshold (θ)
  - $\hat{y} = \begin{cases} 1, & \text{if } \sum x_i \geq \theta \text{ and no inhibitory input active} \\ 0, & \text{otherwise} \end{cases}$
- Key Characteristics
  - Binary inputs and outputs
  - No learning (weights are fixed)
  - Threshold-based decision
  - Inhibitory inputs override excitatory ones

pra-sami

33

MP Neuron



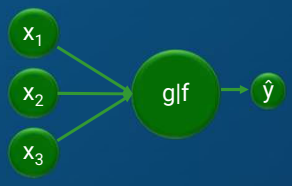
id	Dry Weather	Low Temp	Homework Done	Team Members	Equipment	Ground	Sum	Played
1	1	1	1	1	0	1	5	1
2	1	1	1	1	1	1	6	1
3	1	1	1	1	1	1	6	1
4	0	1	0	1	1	1	4	0
5	0	0	1	1	1	0	3	0
6	0	0	0	0	0	1	1	0

5/2/2026

pra-sâmi

34

MP Neuron



id	Dry Weather	Low Temp	Homework Done	Team Members	Equipment	Ground	Sum	Played
1	1	1	1	1	0	1	5	1
2	1	1	1	1	1	1	6	1
3	1	1	1	1	1	1	6	1
4	0	1	0	1	1	1	4	0
5	0	0	1	1	1	0	3	0
6	0	0	0	0	0	1	1	0

The logic is straight forward. Let's implement this model on a dataset.

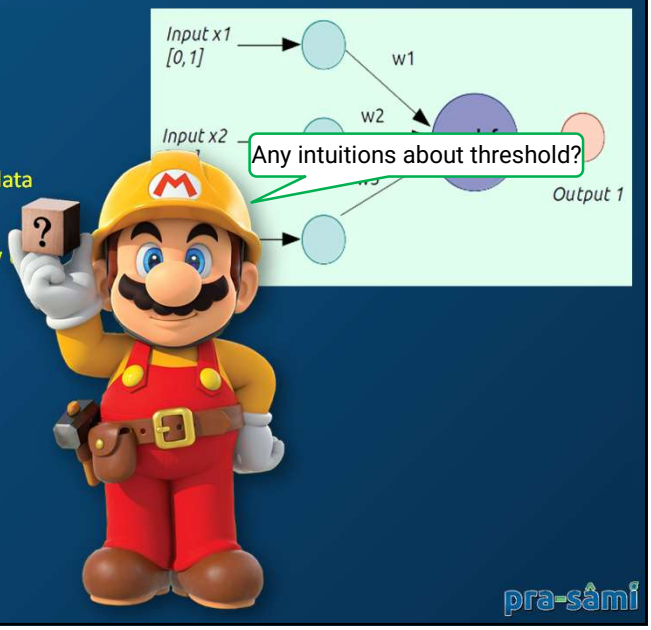
5/2/2026

pra-sâmi

35

Code Example1 – MP Neurons

- Need a dataset with plenty of features and binary output
- Load modified Breast Cancer dataset
  - This dataset is based on scikit-learn breast cancer data
- Its features are a continuous and we need binary
  - For b in range [0, num_features+1]
    - Sum it by row and compare with b
- Converted file is in the shared folder



5/2/2026

pra-sami

36

Next Session

- Perceptron
- Single Layer Neural Network
- Overview of back propagation of errors

5/2/2026

pra-sami

37



5/2/2026

pra-sâmi

ADDITIONAL MATERIAL

pra-sâmi

39

Biological Neural Nets

- Pigeons as art experts (Watanabe et al. 1995)

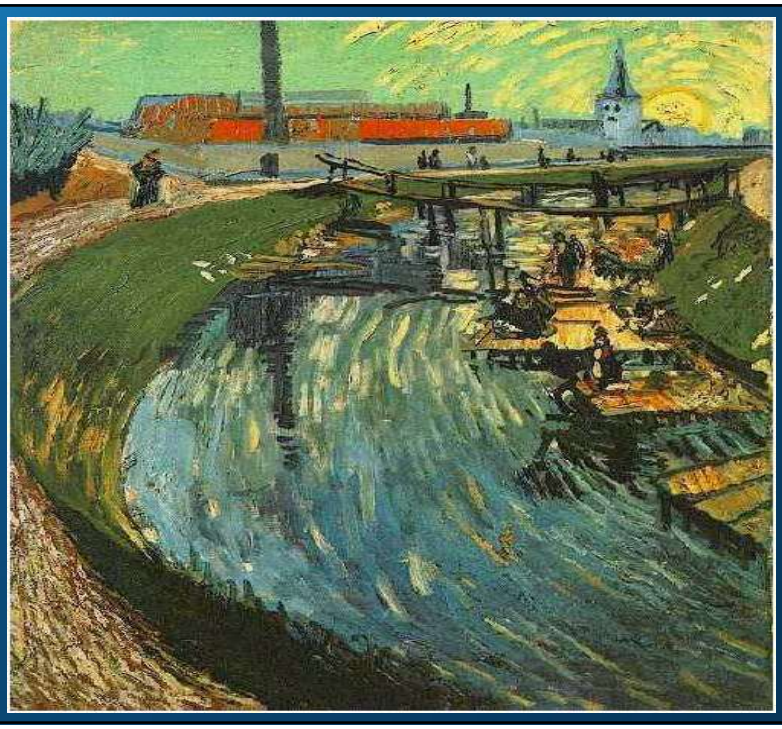
- Experiment:
  - Pigeon in Skinner box
  - Present paintings of two different artists (e.g. Chagall / Van Gogh)
  - Reward for pecking when presented a particular artist (e.g. Van Gogh)



5/2/2026

pra-sâmi

40



5/2/2026

pra-sâmi



pra-sâmi

42 Biological Neural Nets

- Pigeons were able to discriminate between Van Gogh and Chagall
  - With 95% accuracy on train set (when presented with pictures they had been trained on)
  - Discrimination, still 85% successful for previously unseen paintings of the artists

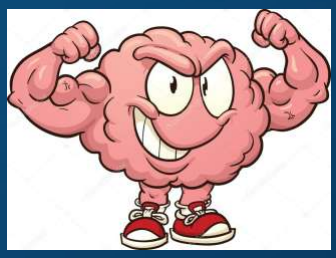
- Pigeons do not simply memorise the pictures
- They can extract and recognise patterns (the 'style')
- They generalise from the already seen to make predictions

- This is what neural networks (biological and artificial) are good at (unlike conventional computer)

pra-sâmi

43 Brain and Machine

- The Brain
  - Pattern Recognition
  - Association
  - Complexity
  - Noise Tolerance



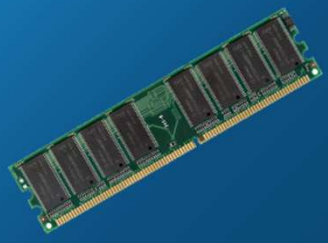


- The Machine
  - Calculation
  - Precision
  - Logic

5/2/2026

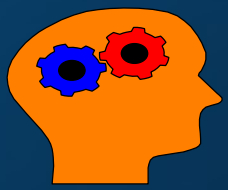
pra-sami

44 The contrast in architecture



- The Von Neumann architecture uses a single processing unit;
  - Tens of millions of operations per second
  - Absolute arithmetic precision

- The brain uses many slow unreliable processors acting in parallel



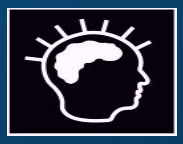
5/2/2026

pra-sami

45

The biological inspiration

- Features of the Brain
  - Ten billion (10^{10}) neurons
  - On average, several thousand connections
  - Hundreds of operations per second
  - Die off frequently (never replaced)
  - Compensates for problems by massive parallelism





- The brain has been extensively studied by scientists
- Vast complexity prevents all but rudimentary understanding
- Even the behavior of an individual neuron is extremely complex
- Single “percepts” distributed among many neurons
- Localized parts of the brain are responsible for certain well-defined functions (e.g. vision, motion).